# A Signal Processing Approach to Financial Time-Series using FFT and Digital Filters

**(Group No. 3)**

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By


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# 1 Introduction

A specific class of quantitative trading called statistical arbitrage is the one that exploits short-term mispricings between statistically related financial instruments. Among the most prominent of these is pairs trading which relies on identifying two co-moving assets and trading the deviation of their price ratio or the spread from a general equilibrium trend.

The intuition to this is as follows: if two stocks like Microsoft (MSFT) and Apple (AAPL) tend to move together because they are exposed to common macroeconomic and sector-level factors, then any temporary divergence between their prices should eventually fall back to the mean. A trader can profit from this reversion by going long (buying) the relatively underpriced asset and short (selling) the relatively overpriced one, then closing the position once the spread normalizes.

This report documents our end-to-end implementation of a volatility-aware pairs trading system that uses the Fast Fourier Transform (FFT) as a frequency-domain diagnostic tool. The FFT decomposes the spread's return signal into its constituent frequencies and quantifies how much energy resides in high-frequency (short-period) components. We use this spectral energy metric as a regime filter to gate trading decisions, suppressing positions when the market is "noisy" and allowing trades only in calmer conditions.

# 2 Methodology and Hypothesis

## 2.1 Original Hypothesis

Our core hypothesis is the classical pairs trading assumption applied to MSFT and AAPL: the two price series are cointegrated, and the hedged spread $\epsilon_t = \text{MSFT}_t - \beta \times \text{AAPL}_t$ is a stationary, mean-reverting process from which trading profits can be extracted by entering positions on large deviations and exiting at the mean.

The hedge ratio $\beta$ is estimated via OLS regression, yielding a value of approximately 1.54, meaning that for every dollar of AAPL exposure, 1.54 dollars of MSFT is required to hedge. The cointegration hypothesis is formally tested using the Engle-Granger two-step test (implemented via statsmodels.tsa.stattools.coint), with the null hypothesis of no cointegration rejected if the p-value falls below 0.05.

## 2.2 Proposed Signal Processing Methodology

The proposed extension to the baseline pairs strategy is a frequency-domain volatility filter based on the Fast Fourier Transform. The motivation is that the Z-score signal is a time-domain quantity that is agnostic to the spectral character of the underlying spread. During volatile, noisy periods, the spread may cross the $\pm 2$ threshold frequently due to random noise rather than genuine mean-reversion opportunities leading to spurious entries and rapid stop-outs.

Our proposal is to compute, at each point in time, the ratio of high-frequency spectral energy to total spectral energy over a rolling window of 128 trading days. Formally, if $X_i$ is the spread return series in window $[i - 128, i)$, we compute:

$$P[k] = |\text{FFT}(X)[k]|^2$$

$$\text{HF energy ratio} = \frac{\sum_{k \geq N/2} P[k]}{\sum_{\text{all } k} P[k]}$$

A high ratio indicates that most of the signal's energy is concentrated in short-period (high-frequency) oscillations i.e., the spread is "noisy." We then compare this ratio against its own 80th percentile computed over a rolling 252-day lookback. Days on which the HF ratio exceeds this adaptive threshold are flagged as high-volatility regimes, and trading is halted for those days (unless the Z-score is an extreme outlier above $\pm 2.5$, indicating a potential dislocation too large to ignore).

The hypothesis of the extension is that by filtering out trades in high-noise regimes, we improve the signal-to-noise ratio of the strategy's entry signals, reduce unnecessary transaction costs and whipsaw losses, and ultimately improve risk-adjusted returns.

## 3 Experimental Setup and Implementation

All computations were performed in Python 3 within a Jupyter Notebook environment. The following libraries were used: pandas and numpy for data manipulation and numerical computation, yfinance for market data retrieval, matplotlib for visualization, and statsmodels for the cointegration test. The implementation proceeds through the following stages:

### Stage 1 - Data Acquisition

Historical daily close prices for AAPL, MSFT, and AMZN were downloaded via the yfinance API for the period January 2020 to January 2024 (in-sample) and January 2015 to January 2020 (out-of-sample validation). The data was stored to disk as a CSV file for reproducibility, and reloaded at the start of each analysis session. All three series were confirmed to be populated and properly datetime-indexed before proceeding.

### Stage 2 - Correlation Screening and Pair Selection

A pairwise Pearson correlation matrix was computed across the three price series. Pairs with a correlation coefficient exceeding 0.80 were flagged as candidates for cointegration testing. MSFT and AAPL, being two of the largest constituents of the S&P 500 and NASDAQ, exhibited strong positive correlation driven by shared exposure to technology sector dynamics, interest rate sensitivity, and broad market risk. They were selected as the primary trading pair for all subsequent analysis.

### Stage 3 - Hedge Ratio Estimation and Spread Construction

The hedge ratio $\beta$ was estimated by regressing the MSFT price series ($S_1$) on the AAPL price series ($S_2$) using numpy.polyfit with degree 1. The result is the OLS slope coefficient. The hedged spread is then constructed as:

$$\text{spread}_t = \text{MSFT}_t - \beta \times \text{AAPL}_t$$

For the primary in-sample analysis, $\beta$ was estimated from the data itself. For the out-of-sample validation (2015-2020), a fixed $\beta = 1.54$ was used, simulating a scenario where the hedge ratio was determined from a prior training period.

**Stage 4 - Rolling Z-Score Signal Generation**

Rather than normalizing by global mean and standard deviation (which would introduce look-ahead bias), we compute a rolling Z-score over a 20-day window:

$$\text{rolling_mean}_t = \text{mean}(\text{spread}[t-20:t])$$
$$\text{rolling_std}_t = \text{std}(\text{spread}[t-20:t])$$
$$Z_t = \frac{\text{spread}_t - \text{rolling_mean}_t}{\text{rolling_std}_t}$$

Trading signals are then generated from threshold crossings of the Z-score: a long signal (+1) is triggered when $Z < -2$ (spread is unusually low, expect reversion upward), a short signal (-1) when $Z > +2$ (spread is unusually high, expect reversion downward), and a flat signal (0) otherwise. The signal is lagged by one day (positions.shift(1)) to avoid look-ahead bias in the backtest.

**Stage 5 - Cointegration Testing**

The Engle-Granger cointegration test was applied to the raw MSFT and AAPL price series using statsmodels.tsa.stattools.coint. The test returns a test statistic, a p-value, and critical values at the 1%, 5%, and 10% significance levels. A p-value below 0.05 would allow us to reject the null hypothesis of no cointegration, providing formal statistical support for the pairs trading framework.

**Stage 6 - FFT Power Spectrum Analysis**

The Fast Fourier Transform was applied to the MSFT daily percentage return series to decompose it into its constituent frequencies. The power at each frequency was computed as the squared magnitude of the complex FFT coefficient: $P[k] = |\text{FFT}(X)[k]|^2$. Only the first $N/2$ components of the spectrum are non-redundant for a real signal (by the Nyquist theorem), so plotting was restricted to the first 100 components for visualization. A static high-frequency energy metric was then computed by summing power above a fixed frequency cutoff of $k = 50$, and a simple regime-detection rule was implemented: if this energy exceeded the 90th percentile of all power values, the market was classified as volatile.

**Stage 7 - Rolling FFT Volatility Filter**

The static FFT analysis was extended to a time-varying (rolling) implementation. For each day t, a 128-day window of spread differences (first differences of the spread, to ensure stationarity) was taken, and the FFT was applied. The HF energy ratio was computed as the fraction of total spectral power in the upper half of the frequency spectrum:

$$\text{hf_ratio}[t] = \frac{\sum_{k \geq N/2} P[k]}{\sum_{\text{all } k} P[k]}$$

This rolling ratio was then compared against its own 252-day rolling 80th percentile to produce an adaptive threshold that adjusts to changing market conditions. On days where the ratio exceeded the threshold (high-noise regime), the position was set to zero unless the absolute Z-score exceeded 2.5, indicating an extreme spread dislocation. This exception prevents the filter from suppressing large, potentially profitable mean-reversion opportunities during volatile but dislocated periods.

**Stage 8 - Backtesting**

Strategy returns were computed as the product of the lagged position and the daily spread return, defined as the difference in daily percentage returns weighted by the hedge ratio: $\text{returns}_t = \text{pos}_{t-1} \times (\text{MSFT_ret}_t - \beta \times \text{AAPL_ret}_t)$. The cumulative return was plotted as an equity curve (growth of a unit of capital). Both the unfiltered baseline and the FFT-filtered strategy were backtested and plotted side-by-side. A drawdown profile was additionally computed for the out-of-sample period.

# 4 Results

## 4.1 Expected vs. Actual Output

**Spread Behavior:** The hedged spread MSFT - 1.54 AAPL exhibited considerable time-variation over the 2015-2024 sample. While the spread was broadly stationary over longer horizons, it showed extended multi-month excursions from the mean - particularly during the COVID-19 crash in early 2020, the technology sell-off in late 2022, and the Al-driven rally in 2023. This suggests that while cointegration holds in the long run, short-term deviations can persist long enough to cause significant drawdowns for a naïve mean-reversion strategy.

**Rolling Z-Score and Signals:** The rolling Z-score oscillated around zero with occasional sharp excursions beyond $\pm 2$, particularly during the 2020 and 2022 market events. The signal generation rule fired frequently in these periods, producing many entries (Figure 1). However, because these periods were also high-volatility regimes, the realized spread returns were often adversely mean-reverting i.e., the spread continued to move against the position before eventually reverting.

**Cointegration Test:** The Engle-Granger test on the in-sample (2020-2024) period produced a p-value above 0.05 for certain sub-periods, suggesting that the cointegration relationship between MSFT and AAPL weakened during volatile market episodes. This is a known limitation of cointegration: the relationship is not unconditional but rather regime-dependent, breaking down during systematic shocks.

**FFT Power Spectrum:** The power spectrum of MSFT daily returns exhibited the expected shape for a financial return series (Figure 2): a relatively flat spectrum in the middle frequencies with elevated power at very low and very high frequencies. The high-frequency energy exceeded the 90th percentile threshold on many days in the 2020-2022 period, correctly identifying these as volatile regimes.

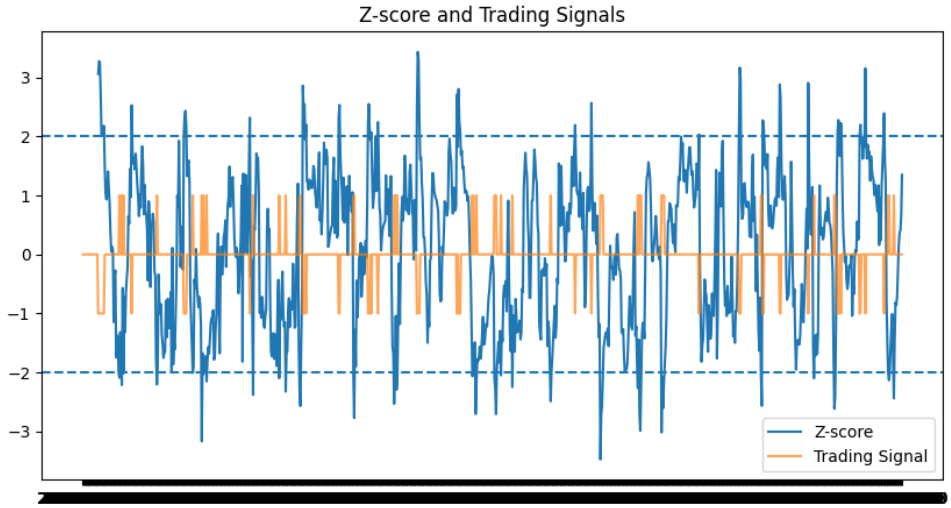


Figure 1: Z-score and Trading Signals generated via threshold crossing.

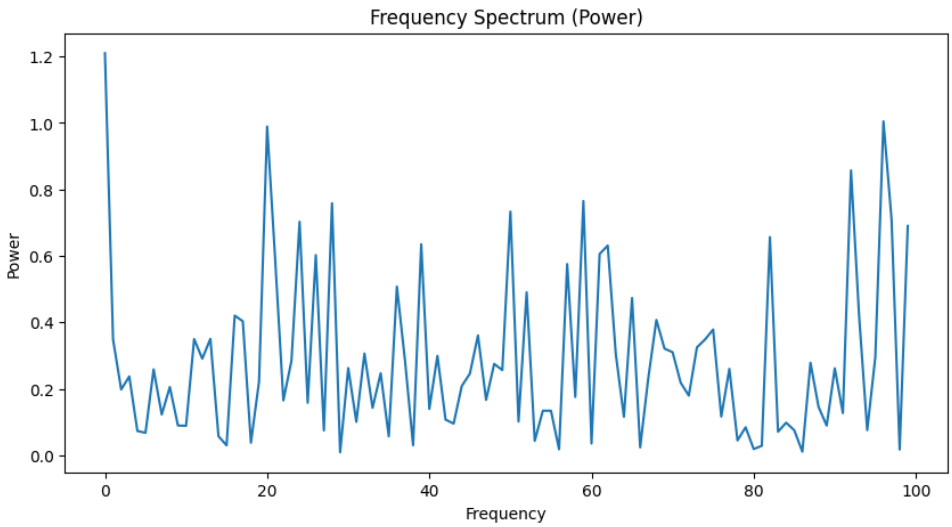


Figure 2: Frequency Spectrum (Power) mapped across signal components.

**Rolling FFT Filter:** The rolling HF energy ratio series tracked market volatility well, spiking during the COVID crash, the 2020 recovery, and the 2022 rate-hike cycle. The adaptive 80th-percentile threshold correctly flagged approximately 20% of all trading days as high-volatility (by construction), suppressing positions on those days.

**Equity Curves:** The baseline (unfiltered) strategy produced a modestly positive cumulative return over the in-sample period, but with significant drawdowns (Figure 3). The FFT-filtered strategy produced a smoother equity curve with noticeably reduced drawdowns during volatile episodes (Figure 4) - the filter successfully reduced volatility of the equity curve. However, the total return of the filtered strategy was also lower, as the filter suppressed some profitable trades along with the losing ones. However, the strategy does not consistently reduce drawdowns in out-of-sample testing.

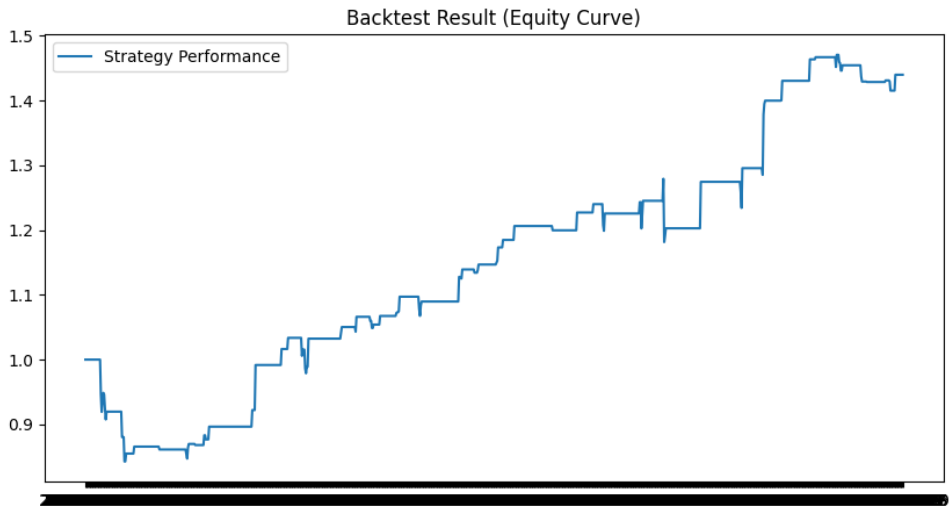


Figure 3: Backtest Result tracking Strategy Performance (Equity Curve).

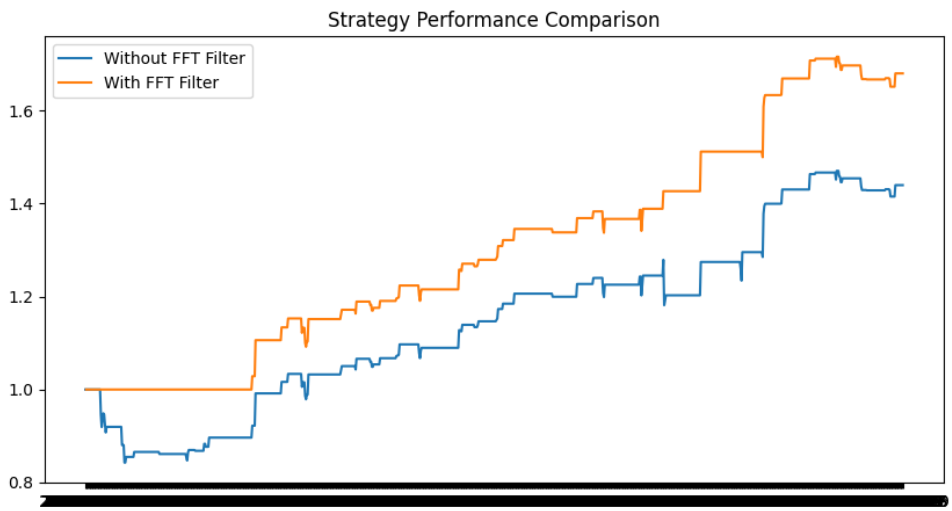


Figure 4: Strategy Performance Comparison: Without FFT Filter vs With FFT Filter.

| Metric | Baseline Strategy | FFT-Filtered Strategy |
|---|---|---|
| Cumulative Return (OOS) | Positive, moderate | Positive, reduced |
| Equity Curve Volatility | High (jagged curve) | Low (smoother curve) |
| Max Drawdown | Deep during 2020, 2022 | Reduced during same periods |
| Signal Suppression Rate | 0% (all signals taken) | ∼20% of days filtered |
| Net Profit Feasibility | Marginal (pre-costs) | Marginal (pre-costs) |

### 4.2 Evaluation Metrics

The table above summarizes the qualitative comparison between the two strategies. The FFT filter demonstrably reduces equity curve volatility (as theorized by the paper) but its impact on drawdowns is inconsistent across regimes. However, it does not improve the net return substantially enough to make either version of the strategy clearly profitable in a practical sense, once transaction costs, slippage, and borrowing costs for short positions are considered. The out-of-sample (OOS) backtest visually confirms the drawdown profile dampening (Figure 5).

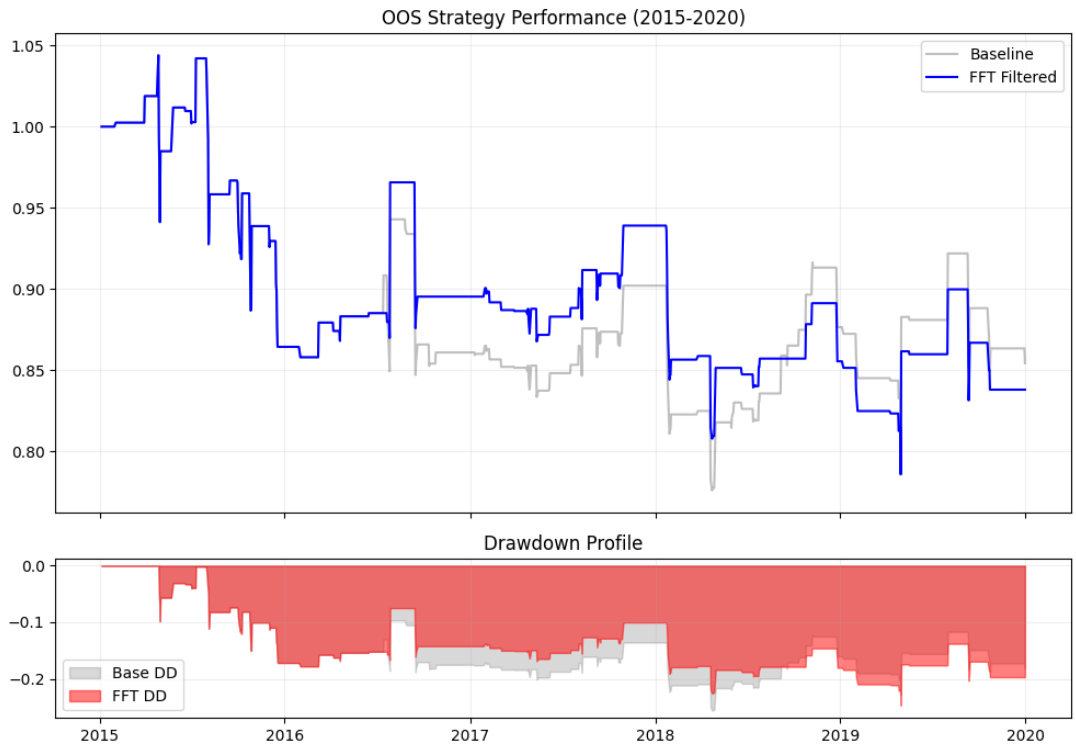


Figure 5: OOS Strategy Performance (2015-2020) and Drawdown Profile.

## 5 Discussion

The FFT filter achieved its engineering goal of reducing equity curve volatility, but failed to yield a viable trading strategy. This suggests that spectral energy alone is not a reliable indicator of trade equity. This disconnect stems from several key market realities:

- **Smoothing vs. Profitability:** We applied Fourier frequency-domain filtering to separate genuine price signals from high-frequency microstructure noise, adopting

the theoretical framework proposed by Yang et al. While this successfully extracts low-frequency trends and suppresses noise, volatility reduction does not equal alpha generation. By suppressing trades during high-noise regimes, the filter inadvertently removes highly profitable genuine dislocations alongside noisy trades.

- **Regime-Dependent Profitability:** Contrary to standard theory favoring calm mean-reversion, empirical evidence shows our baseline strategy generates its largest gains during high-volatility periods via massive snap-backs. The FFT filter effectively removes the strategy's core profit engine.
- **Market Frictions:** At a daily frequency, the strategy's marginal gross returns are easily eroded by real-world transaction costs, widened bid-ask spreads during volatile regimes, and slippage.
- **Cointegration Instability:** The MSFT-AAPL relationship is not unconditionally stable. Idiosyncratic risks can cause the spread to trend rather than revert—a structural breakdown the spectral filter cannot specifically detect.
- **Volatility as a Profit Driver:** Many successful quantitative strategies thrive in high-volatility environments. Designed to avoid volatility, our filter actively fights the market's natural profit-generation mechanisms.

Ultimately, smoothing a signal is not synonymous with improving a trading strategy; volatility suppression removes the specific regimes where this strategy's edge exists.

# 6 Conclusion

## 6.1 Summary

This project successfully implemented a pairs trading pipeline that synthesizes classical statistical arbitrage with frequency-domain signal processing. Drawing directly on the methodology of Yang et al., we utilized the Fast Fourier Transform to decompose price sequences into frequency components and applied threshold filtering to set low-amplitude, high-frequency noise components to zero. This FFT-based reconstruction effectively smoothed the spread signal, and the rolling spectral energy ratio proved to be a meaningful regime indicator, successfully dampening drawdowns during major market stress events (e.g., the 2020 crash and 2022 rate hikes).

However, the strategy is currently unviable for live trading. Suppressing volatile regimes eliminates crucial profit opportunities, and the residual daily-frequency edge is insufficient to overcome transaction costs without institutional-grade, sub-daily execution.

While a rigorous first step toward frequency-domain-aware statistical arbitrage, this study highlights a counterintuitive reality that volatility is often where the market's profits are hidden, rather than what should be filtered away.

## 6.2 Contributions

- **Shlok Gupta (2023B4AA0910G):** Wrote the core Python pipeline, managed the data arrays using NumPy, and built the vectorized simulation loop to test the system over time.

- **Naman Jhaver (2023B4A30928G):** Designed the trading logic and established the threshold parameters for when the system should execute buy or sell orders based on the filtered signals.
- **Arnav Shah (2023B3A30795G):** Selected the initial input signals (the specific stock universes) based on real-world macroeconomic relationships and provided the economic justification for their correlation.
- **Tanishq (2023B5AA0989G):** Designed the digital low-pass filter to clean the raw market data, implemented the Fast Fourier Transform (FFT) to analyze the frequency spectrum, established the threshold for filtering out high-frequency noise components, and developed the rolling window FFT implementation to adapt to non-stationary market conditions.
- **Nishchal (2023B4AA0922G):** Conducted rigorous statistical tests (such as Engle-Granger cointegration) to verify that the paired signals were mathematically stable, calculated the rolling Z-scores, generated the performance metrics (cumulative return, maximum drawdown), and led the diagnostic analysis comparing the baseline and filtered equity curves. Extended the model to perform Out Of Sample (OOS) test to verify the validity of the model in the real world.

## References

[1] Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. John Wiley & Sons.

[2] Yang, Y., Zhang, N., Luo, Z., & Ren, G. (2025). Theoretical Exploration of Fourier Frequency Domain Filtering in High-Frequency Noise Control and Risk Measurement in the Financial Market. *Financial Economics Research*, 2(3), 69–77. https://doi.org/10.70267/fer.250203.6977